ChemRMBench: Evaluating Event Memory and Operation Timing in Laboratory Agents
Native Evidence and a Preserved Task-Clock Contract

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Abstract

Laboratory agents must retain past events and perform the corresponding operation at the required effective time. A correct final object pose does not establish temporal compliance. This addendum documents full registered-state archives at eight task boundaries, one validated full-world restart boundary, explicit agent-authored memory and measured contact intervals. A naive continuous case returns the bottle but contacts at 15.413–15.417 seconds for a 10-second target. A separate expiry case refuses without executing any tail commands. A new native case retains the original lift event and 30-second contact target across an interrupted child and cold retry: the failed attempt consumes 2.3875 native seconds, and restoration in both children is charged, giving a sampled timing-error interval of $[-4.165, +4.168]$ ms. Its logical simulation clock excludes fresh-scene initialization and process downtime. A separate three-branch test holds the prior proposal and delivery debit fixed, confirming cold continuation, feasible retry and zero-tail refusal after dispatch-window expiry. These are inspectable engineering witnesses, not causal memory gains, calibrated chemistry or a complete benchmark.

1. Question and scope

An agent may recognize a laboratory task while forgetting which event occurred earlier or when it occurred. Conversely, an agent may recall a deadline correctly but execute late because a model call or a stage reconstruction consumes unaccounted time. The central evaluation question is whether the *correct operation occurs at its required effective time given the permitted history*. It is not answered by a terminal placement predicate alone.

Existing work establishes the relevant neighboring capabilities. MATTERIX provides a multiscale laboratory twin, hierarchical workflows and a modular semantics engine (Darvish et al., 2025). RoboMME studies robotic temporal, spatial, object and procedural memory (Dai et al., 2026); RoboMemArena adds long trajectories and memory-related annotations with recent and keyframe buffers (Lei et al., 2026). ReflexBench decouples simulation stepping from robot control and supports synchronous and asynchronous latency evaluation (Chen et al., 2026). Thus neither robotic memory, laboratory workflows nor continued evolution during inference is novel by itself. The candidate ChemRMBench contribution is its frozen task/input/evidence contract and independently checkable joint compliance. Matched scientific comparisons are still required.

This document extends the existing Paper A measurement-instrument draft with new native engineering evidence. Historical symbolic process studies, synthetic readout fits and native observation audits remain separate cohorts. The new bright-cue bottle cases do not repair their weak observation controls. Policy learning and adaptive-policy claims remain outside this addendum.

2. Measurement and continuation contract

2.1. Events, clocks and admissible claims

For effective operation events A and B, target gap D and tolerance ϵ, define

$$e_T = t_B^{\text{event}} - t_A^{\text{event}} - D. \tag{1}$$

Commands, image observation times and effective physical events are distinct. With conservatively closed sampled intervals $A = [a^l, a^u]$ and $B = [b^l, b^u]$ on the same declared clock,

$$E_T = [b^l - a^u - D,\ b^u - a^l - D]. \tag{2}$$

Timing passes only if the *entire* interval lies in $[-\epsilon, \epsilon]$. These brackets express sample resolution, not confidence intervals. Missing or censored event evidence is unavailable, not a successful never-acted inference. Native epochs and host

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Research progress addendum. Native engineering evidence; not submitted.

clocks cannot be subtracted without an explicit alignment model.

The V83–V87 contact pilots use a declared *new child resume origin* as their anchor. V90–V92 preserve the original event anchor on the logical task clock in Section 4. Neither cohort measures an uninterrupted chemical $A \to B$ interval. Contact is a native bilateral-grasp event, with 240 Hz sampling and a three-sample confirmation rule. Its predicate requires finger-associated contacts on both sides with impulse norm above 10^{-8} in the recorded native units; physical-unit calibration remains unverified. Its independent sensor/contact calibration is not established. The short cue is a noncolliding visual fixture, not a chemical event.

2.2. Material state and explicit memory

The preserved parent executes 7,920 pinned drive-target commands and seals five stages at a released bottle on the scale. A fresh native process reconstructs 56 rigid bodies and nine articulations. All 65 entities must satisfy the original 1 mm, 0.5° and 10^{-4} rad restoration limits at each of 30 transient samples. Initial restored root/joint velocities must differ by at most 10^{-6} in their recorded units; the live task separately limits arm speed to 30 rad/s and finger speed to 2 m/s. Hidden PhysX solver state is not restored. Transport limits, 20 mm relative displacement and 80% bilateral contact, are distinct from restoration limits.

The physical checkpoint is bound to the fixed controller cursor, original return goal and an agent-authored external semantic-memory record. Restored execution starts at command 7,920 and runs only the 5,507-command tail. For V83–V90 readouts, the actor receives a current image, joints and declared public metadata plus retained memory or a cleared condition. V91 and the V92 archive extension instead hold an archived parent packet and its prior model proposal fixed, as detailed in Section 4.3. Old cue pixels are not attached to these readouts. This restores explicit records, not hidden model state. No object pose or robot joint-state writes occur during the live tail after declared initialization/restoration; movement follows the fixed drive-target primitive and new native integration.

The requested client model is `gpt-6-astra`; the provider's backend checkpoint identifier is not exposed. Each recorded completed call is checked for tool use and token usage. A successful model proposal is executed only after schema, identity and time checks. A declined or late proposal has no scripted substitute.

2.3. Readout time and refusal

In the historical V83–V84 paused cases, both retained and cleared readouts occur without physics callbacks. In the historical V85–V87 continuous cases, each observed integration callback during the retained readout advances native time. If the proposed dispatch uses tick n_d and readout consumes n_r ticks, the executor waits $n_d - n_r$ further ticks and refuses if that budget is negative. It never silently clamps a late proposal to an immediate action.

Table 1. Independently recomputed native cases. D and contact are seconds after each new resume origin. Timing tolerance is 20 ms; return-position tolerance is 20 mm, separate from the full transport gate. V87's refused attempt fails the task with zero tail commands; its retry PASS uses a new origin.

Case	D	Contact (s)	Return (mm)	Timing
V83 amber	16	16.000–16.004	0.330	PASS
V84 green	10	10.000–10.004	0.330	PASS
V85 accounted	10	10.000–10.004	0.406	PASS
V86 naive	10	15.413–15.417	0.444	FAIL
V87 refused	10	unavailable	unavailable	REFUSED
V87 retry	10	10.000–10.004	0.406	PASS

The historical continuous cleared readout is one offline diagnostic on the archived identical current packet after the native worker closes. It is excluded from native readout latency and is not physically executed. Native and host elapsed times are recorded separately; no unit realtime factor is assumed. The naive comparison deliberately waits the original n_d ticks *after* readout, in isolated simulation evaluation only.

3. Native engineering cases

Table 1 and Figure 2 distinguish timing from final placement. All cases use one scene and one fixed primitive. They are not independent random repetitions or a paired success-rate estimate.

The source-bound configuration uses Isaac Sim 5.1.0 and Isaac Lab distribution 0.54.2 with 240 Hz CPU physics. The fixed command archive has SHA-256 prefix `03a63a0b77037785`; the V87 executor seal has prefix `4509d3f515765954`. Per-case configurations and recomputation receipts are indexed in `figure_sources.json` beside the figure source. The prior V80 contact trace supplies a nominal dispatch-to-contact duration of 5.172916936 s, not a calibrated prospective bound. Dispatch rounds upward to the next native command boundary. Confirmation occurs after three positive contact samples; the reported interval brackets the first positive sample, rather than the later confirmation time.

V83 and V84 pause native physics during readout. V85 spends 67 native readout ticks (0.279167 native seconds versus 12.840302 host seconds) once and then waits 1,092 remaining ticks. V86 spends 1,299 readout ticks (5.413 seconds), then erroneously waits another full 1,159 ticks. Its contact error is 5,412.501–5,416.667 ms; subtracting the



Figure 1. Unedited source RGB frames from V87. Left: the parent's short cue observation. Middle: the failed child's current observation *before* its model response, not a new photograph after refusal. Right: the completed retry tail. These frames come from the preserved laboratory scene; they are not newly synthesized scene illustrations.

measured readout duration leaves the same 0.001–4.167 ms sampled skill residual. Bottle return still passes. This within-case arithmetic supports the bookkeeping explanation; different model/native readout delays prevent a matched causal comparison between V85 and V86.

In V87, the first actual native child consumes 58 model-readout ticks plus a predeclared 1,440-tick delivery-delay injection. The six injected native seconds are *not* measured model latency. It refuses before dispatch and executes zero tail commands. All 45 parent task files, the source checkpoint and explicit memory remain unchanged. One fresh child retry then executes only 5,507 tail commands and meets the new-attempt contact target. The first refusal remains a failed task attempt. A preceding pre-SDK free-memory-gate rejection is preserved separately; it enters no native simulation and causes no repeated parent or memory ingestion.

V84–V87 use 145,010 observed input-plus-output tokens in 13 completed client calls and at most 0.528 allocated GPU hours for owned native workers. This worker-wall-time bound is not a measured GPU-utilization integral and excludes provider compute. No new learned policy is fitted in these pilots. V87 alone uses four calls, 44,816 tokens and a 0.148-hour native-worker allocation bound.

4. Preserving the original event deadline

4.1. Committed clock and native integration

The old material checkpoint has no native timestamp or pending timer field. Therefore its successful reconstruction does not establish deadline preservation. The new CPU component binds an immutable event/timer header to a physical checkpoint and an ordered attempt ledger. For checkpoint task time τ_C, completed native attempts $j < k$ and a fresh native epoch k, its explicit logical simulation clock is

$$\tau_k(t) = \tau_C + \sum_{j<k}(t_j^{\text{end}} - t_j^{\text{start}}) + (t - t_k^{\text{start}}). \quad (3)$$

Table 2. V90 task-clock treatment. Excluded phases still consume host or native setup time.

Phase	Task clock
Scene construction and 1,200 warmup steps	Paused
Post-checkpoint memory ingestion	Paused
Host downtime between children	Paused
30 material-restoration steps per child	Charged
Readout, failure injection, wait and tail	Charged

The declared clock excludes host downtime and fresh-scene construction, including 1,200 warmup callbacks (about five native seconds) inside each child before material restoration. The fresh scene evolves during warmup while the saved task state is logically paused. Every subsequent native restore step, readout, failure injection, wait and action is charged from the instant material restoration begins in the V90 adapter. It logs exact native and logical before/after timestamps for every callback. Old transient snapshots lack these timestamps and remain unchanged. This logical clock does not model chemical evolution during scene construction or host downtime.

Given event interval A and a declared skill-duration estimate $[\eta^l, \eta^u]$, the start times that can satisfy the full interval tolerance lie in

$$\mathcal{W} = [a^u + D - \epsilon - \eta^l,\ a^l + D + \epsilon - \eta^u]. \quad (4)$$

An empty window means event/skill uncertainty is already too large; a reply after its upper endpoint is expired. A time below the lower endpoint permits scheduling but forbids immediate dispatch. The gate is rechecked after waiting at the actual command boundary. The native wrapper reads A, D and tolerance from the committed header and verifies its actual physical/controller/memory binding; a substituted

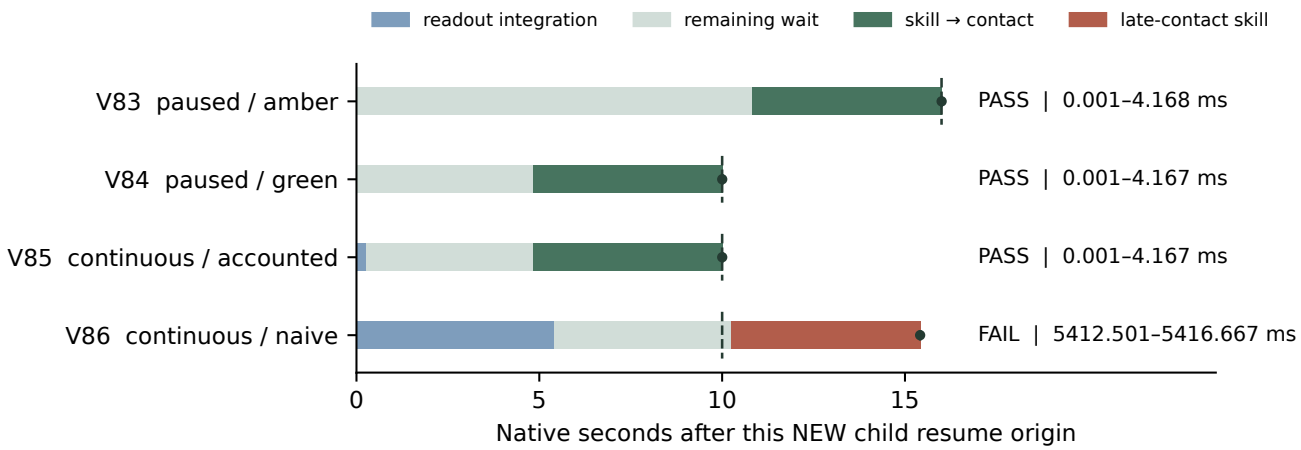


Figure 2. Measured native timing, with the target marked by a dashed line. The 4.167-ms contact bracket collapses to an endpoint dot at this seconds-wide scale; adjacent millisecond ranges show sampled timing error, not statistical uncertainty or exact zero error. Paused calls spend zero native readout time. The naive case charges another complete wait after readout. Unequal delays prevent a matched performance comparison.

event or gap has no fallback. Missing, edited or unfinished predecessors and reused native epochs refuse continuation. Twenty-two CPU checks cover storage/arithmetic, overflow, early dispatch, closed endpoints and the immutable-header gate. Their passing does not establish event calibration. Mapping an interval never grants scientific admission.

4.2. Native original-event recovery witness

V90 seals a new executor and parent. A declared ideal causal simulation instrument reports the 30 mm lift interval, with a short visual cue at its 36-sample confirmation. Astra authors explicit memory from that permitted observation. After five stages, the material checkpoint binds that memory and the original event/timer header. Current child packets contain the current image, joints, logical task time and public skill estimate; the old report and cue pixels are absent. The actuator remains the same fixed drive-target primitive.

The original lift interval, in task seconds, is

$$A = [16.862501, 16.866668], \qquad D = 30.$$

The receipt retains full precision. The parent checkpoint is at 33.000001721 s. The first child advances 30 restoration, 63 model-readout and 480 predeclared interruption ticks, totaling 2.387500125 native seconds. It fails without executing any tail commands while the original start window is still feasible. The fresh retry inherits task time 35.387501846 s, not the old checkpoint time or a new zero. It advances another 30 restoration and 693 model-readout ticks, waits 790 ticks and rechecks the window at dispatch. All 65 entities pass all 30 original restoration gates in both children.

Measured retry contact on the original task clock is $B = [46.862502, 46.866669]$ s. Using full-precision receipt values, Equation 2 gives $E_T = [-4.165102, +4.168232]$ ms, entirely within 20 ms. The retry runs only 5,507 tail commands and returns within 0.456 mm. All 71 parent files, A/D, checkpoint and memory are unchanged. The first task attempt remains failed. Four real calls use 44,853 tokens; native worker allocation is bounded above by 0.138330 GPU hours, excluding provider compute. An independent watchdog limits each native process to 420 s plus 15 s cancellation grace; all three owned groups close. The requested model's backend checkpoint remains unavailable.

The predeclared skill-duration envelope is the prior V80 estimate $5.172916936 \pm 2/240$ s. It is not a calibrated prospective guarantee. Actual contact evidence determines the final interval verdict. Figure 3 is one engineering witness. The next section adds separate controlled native branches; V90 itself has no matched controls. Neither case supplies an average recovery gain or causal memory benefit.

4.3. Matched native executor boundaries

V91 captures a separate sealed parent and tests three predeclared branches from its released checkpoint. Their A/D header, material state, fixed primitive and prior Astra proposal are identical. Two real calls ingest the permitted cue and author the shared proposal after the parent process closes, before all children. Each child then advances exactly 240 native delivery ticks; this is a controlled debit, not measured model latency. The shared readout uses the archived command-7,920 parent RGB, last measured joints and checkpoint task timestamp. Each fresh child observation is recorded but is not sent to the model. Independent ledger copies share the same committed header. The no-fault control is a cold continuation, not a continuously running parent.

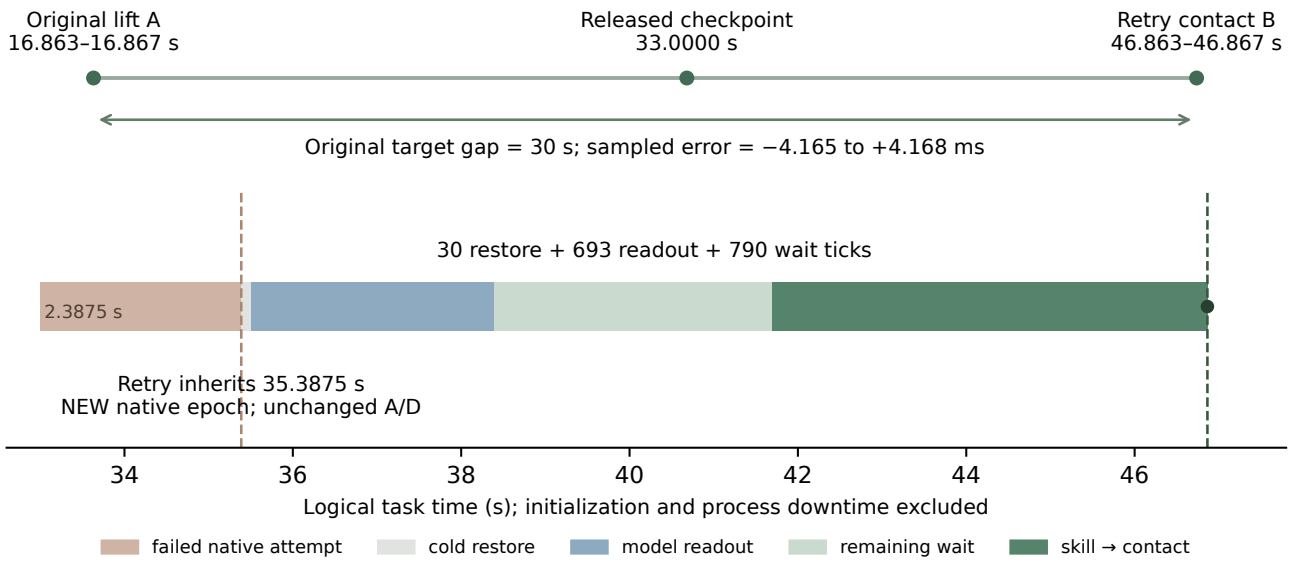


Figure 3. V90 preserves the original event target on the declared logical simulation clock. Above: measured A, released checkpoint and retry contact B; sample brackets collapse at this seconds-wide scale. Below: the failed 2.3875 native seconds remain charged before the retry's restore, readout, remaining wait and skill. Different native epochs share an explicit task-time mapping. Fresh-scene initialization and process downtime are excluded; this is not a chemical-time or realtime deployment result.

Table 3. V91 controlled branches. Errors are relative to the shared original A/D, in ms. Placement tolerance remains 20 mm. These are branches of one parent, not a success-rate sample.

Branch	Outcomes	E_T (ms)	Return (mm)
No fault	Completed	$[-4.165, 4.168]$	0.383
Recovery	Failed, completed	$[-4.165, 4.168]$	0.383
Expired	Refused	unavailable	unavailable

The no-fault branch restores once and completes the tail. The recovery branch first charges 30 restore, 240 delivery and 480 injected interruption ticks, totaling 3.125000163 native seconds, then cold retries from the unchanged material checkpoint at task time 36.125001884 s. Its second restoration is also charged. Both feasible branches contact on the original interval and return within 0.383 mm (Table 3). The expired branch charges 30 restore, 240 delivery and 2,400 injected ticks; it refuses at 44.125002301 task seconds, beyond the original start-window upper endpoint 41.701250610 s. It sends zero tail commands and has no retry. Failure time and outcomes remain recorded.

All four cold restores check 65 entities at every one of 30 transient samples under the original gates. All 71 parent files are unchanged; each successful branch executes only the 5,507-command tail. Five owned native groups close, with a 0.234165 GPU-hour allocation upper bound; the two prior calls use 22,400 tokens. Raw recomputation checks all 121 pinned sources, every callback, both observed contact intervals and the zero-tail refusal. A committed protocol-scope addendum identifies three unused inherited readout labels. The consumed protocol remains unchanged; V91's controlled-delivery fields and actual call records describe execution.

Stage evidence and restart coverage are distinct. In V91, the combined prefix and tail commit all eight stages' trajectory evidence. Only the released scale boundary has a sealed full-world task checkpoint and native restart validation. Other stage manifests contain robot joints and the task object's pose; they do not establish full-scene reconstruction there. The coverage inventory counts one validated task boundary, excluding the setup checkpoint. Mid-carry and liquid restart remain open.

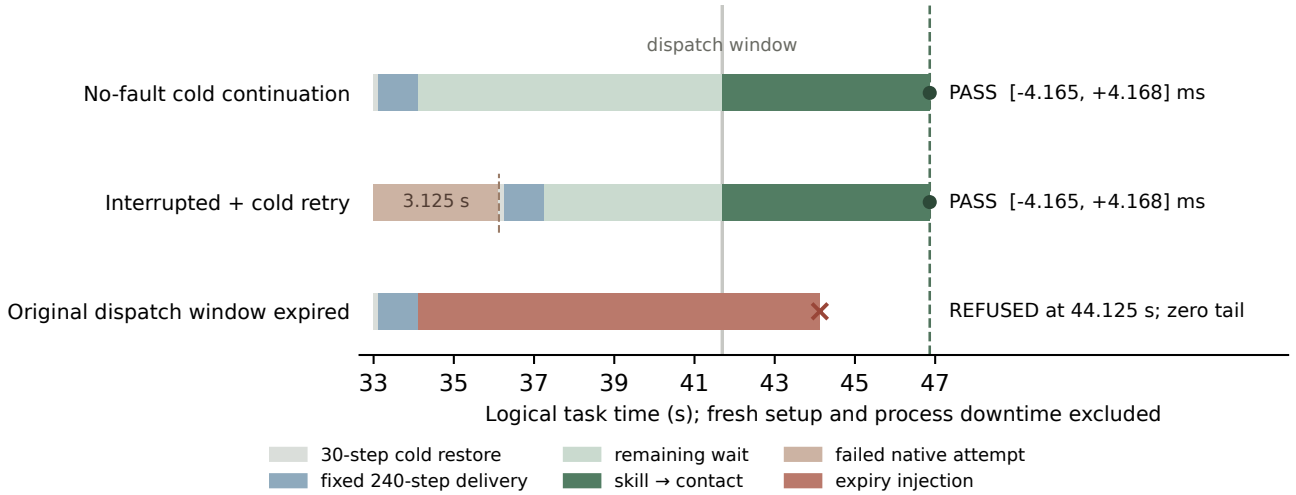


Figure 4. V91 holds the prior agent proposal and delivery debit fixed while changing the native interruption condition. A failed attempt's 3.125 seconds remain charged before retry. The expired branch refuses without tail commands. The shaded dispatch window and sampled contact brackets collapse at this scale; numerical intervals provide the verdict. Bars stop at measured contact for feasible branches, whose full tail continues for 5,507 commands. No-fault is a cold control. Setup/downtime are excluded, and delivery is synthetic controlled advancement.

4.4. Input boundary and next controls

Old CLI working-directory names include cue-condition words such as `green`. The actual provider ambient context was not captured, so those request envelopes do not establish their exclusion from model input. The new client uses opaque temporary directories, neutral filenames and an invocation fence; private archival paths are outside the actor call. V89's two-call archived-input diagnostic uses 22,390 tokens, with retained memory scheduling and cleared memory waiting on the same image. These calls are excluded from all reported native-case totals and add no native timing result. Its frozen client predates the concurrency fence. V90 separately records safe configuration-source and environment-key manifests, removes stale shell working directories and experiment environment fields, and audits all four serialized public requests. Credential values are not archived. Undocumented provider context remains unattested.

The next experiment must add a continuously live parent control, randomized event times, equal-instruction memory interventions and additional tasks/scenes. V91 closes controlled cold-continuation and original dispatch-expiry boundaries; its fixed proposal does not test causal memory effects. V90's ideal event report is a declared observation-resource condition, not independently calibrated sensing. Privileged future contact labels remain outside the actor interface. A rigid event proxy is still not chemistry.

5. Registered-state capture at stage boundaries

The V91 stage manifests contain limited boundary views. A separate V92 executor now archives all registered rigid and articulation state at every command-stage boundary. For each of 56 rigid bodies it saves pose and root velocity; for each of nine articulations it additionally saves joints, velocities, recorded drive targets and available runtime properties. Host contact history, controller graph, original return goal and source identity are retained. A command cursor and a recorder's host tick remain separate coordinates: setup, restoration and waiting can advance one without consuming the next source command.

Each archive binds the sealed stage manifest, source-command cursor and complete registered entity set. An archive commitment is distinct from the restart commitment. Saving a dynamic state does not establish restoration of hidden solver state, contact caches, agent context or pending timers. The executor reads the archive back and compares it exactly with the captured state, captures the live state again and asserts that both simulator time and callback count are unchanged. A separate CPU reader verifies archive hashes and checks robot joints, velocities and drive targets, bottle pose and recent contact history against the raw boundary rows.

One fresh parent collects the first five stages and one cold child collects the final three. All eight archives contain 65 registered entities; all 91 parent files remain unchanged. The released scale boundary still is the only natively validated restart boundary, with 65 entities checked at each of 30 restore samples. Thus the observed state-save coverage is $8/8$, while restart coverage remains $1/8$. Figure 5 shows untouched source frames and both coverage measures. It does not demonstrate restoration at a grasp, transport or liquid stage.

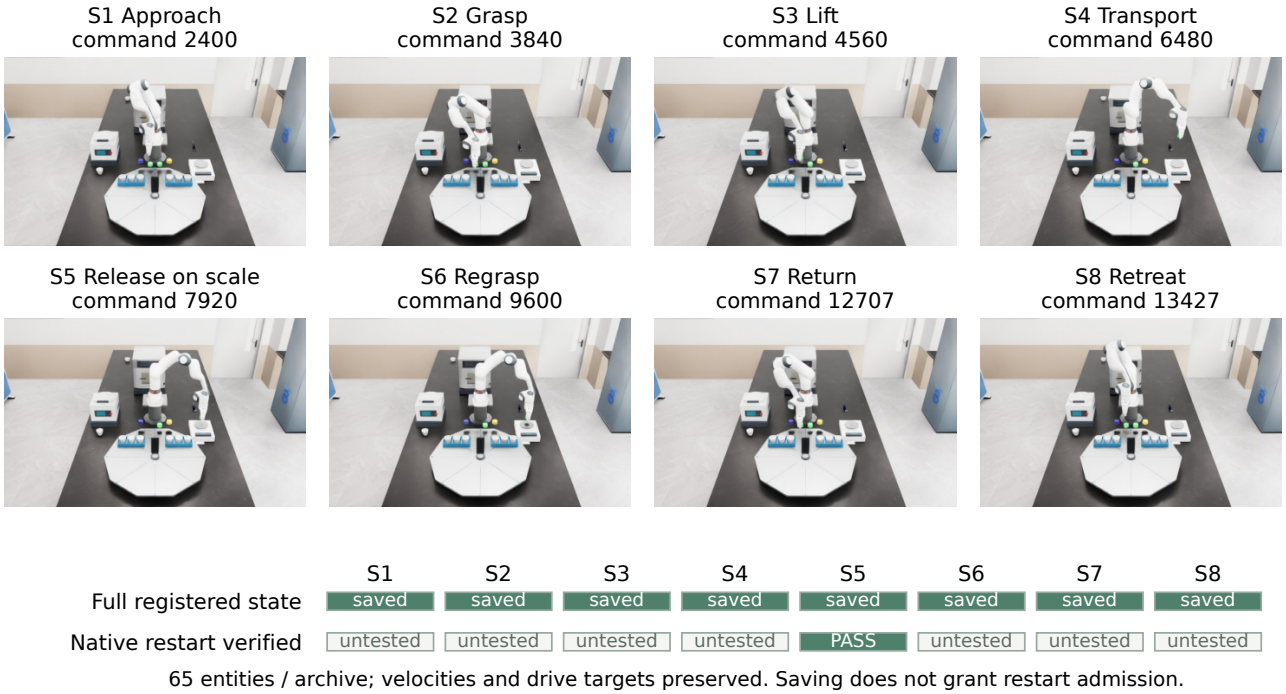


Figure 5. V92 archives registered rigid, articulation and declared host state at all eight source-command boundaries on the preserved laboratory scene. The unedited native RGB frames accompany the saved states. The two coverage rows distinguish successful state persistence from native restart validation: only S5, the released scale boundary, has the latter evidence. Velocities and recorded control targets are preserved; hidden solver, particle and new agent/timer bindings are not restored.

The child continues only the remaining 5,507 commands under the original logical event clock. Its contact error interval is again $[-4.165, +4.168]$ ms, and return error is 0.383 mm. The same prior-proposal/240-tick delivery protocol uses two actual calls and 22,398 tokens; this is not a new memory intervention. A pre-SDK child launch rejects insufficient free GPU memory. That rejection stays archived; one predeclared child-only resource retry adds no parent commands or model calls and does not reset the task clock. The worker allocation bound, including the rejected startup, is 0.115482 GPU hours. A separate resource reader extension changes bookkeeping only, while the originally sealed native and archive sources stay unchanged. The original reader was sealed before collection; its resource extension was sealed after startup refusal and before the child-only resource retry. The current scoped CPU gate has 111 tests and seven subtests, including eight archive-interface tests; this software gate is separate from native or scientific benchmark admission.

6. Limitations and experimental frontier

These are fixed-primitive engineering cases, not a benchmark leaderboard or a learned-memory attribution study. V90's original-deadline witness remains a single parent with two linked child attempts, not independent successful repetitions. V91 likewise has one shared parent and three controlled branches, rather than independent repetitions. V92 has one parent and one linked cold child; eight archived boundaries are not eight independent task episodes or eight validated restarts. The V90 failed child spends 0.262500 native seconds during a 12.976087-second host readout. The retry spends 2.887500 native seconds during a 16.740669-second host readout. These differing clock ratios do not imply a fixed realtime factor. Cleared readouts follow an abstention instruction, so waiting does not establish superiority under equal instructions. Missing randomized matched contexts, possible residual current-observation information, unknown backend checkpoint, ideal instrument observations and uncalibrated contact remain limitations.

Hidden model and solver states, mid-carry reconstruction, particles, liquid restart and real-world recovery are unverified or unsupported. The existing historical cue-control failures remain. Formal M1 admission and qualified new native training episodes are zero. A complete benchmark requires task-graph and observation controls, multiple scene/task instances, matched memory and policy interventions, retained failures and declared statistical units. Formatting this addendum with the established ICML style is not evidence that those research requirements have been met.

Impact statement

The current artifacts improve inspectability of simulated laboratory-agent evaluation. They do not authorize autonomous chemical operation or certify safety. Treating a new attempt as preserved chemical time would hide an irreversible protocol violation; the explicit clock and failure boundaries are intended to make that error detectable before deployment.

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